68746: Advanced Quantum Science

Quiz 3: *Take-home Quiz*

(20 marks = 20%)

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Quiz 3 Instructions

You are expected to submit your own, independent solutions for this assessment. We understand that you will want to collaborate to some degree, and we encourage collaboration in a general way. However, I want to see you present your solution in your own way, and to illustrate your own understanding in your submission. If your solutions are too similar, I will not be able to see that you individually understand the work, and I will not be able to award full marks. Therefore, please bear this in mind as you decide how to collaborate with your colleagues.

You should treat the use of generative AI tools in a similar manner. Be aware that GenAI can easily make mistakes with this kind of material, uses its own conventions which are different from those we have used during the course, and often leaves out crucial steps in reasoning. We do not prohibit the use of GenAI as you work on this assessment piece, but again, the work you submit must be presented in your own way, and illustrate your own individual understanding. If you use GenAI, it should be as a supporting tool, and you are responsible for ensuring the accuracy and clarity of your submitted work. If your submission does not clearly demonstrate that you understand what you have submitted, then I will not be able to award full marks.

Please also bear this advice in mind in the context of working out how much working to include. It is not sufficient to provide enough working so that I can work out that it is correct using my own knowledge of the solution. You need to provide enough working to show me that *you* know and understand why it is correct.

Please submit your work, through Canvas, in electronic format. Both hand-written and electronically typeset solutions are acceptable—so long as your solutions are clear, legible and logical, *and equations and maths are properly formatted*, any reasonable solution will be accepted.

Quiz topic: Dispersive joint two-qubit measurement

In the lectures, we derived the Hamiltonian for two-qubits coupled to a single cavity, but far detuned, so as to be in the dispersive regime relative to the cavity:

$$\frac{H}{\hbar} = \sum_{j=1}^{2} \left[\frac{\Delta_{0,j}}{2} + \chi_j \right] (\sigma_{\mathrm{j,0}} - \sigma_{\mathrm{j,z}}) - \sum_{j=1}^{2} \chi_j a^\dagger a \sigma_{\mathrm{j,z}} + \frac{J}{2} (\sigma_{1,\mathrm{x}}\sigma_{2,\mathrm{x}} + \sigma_{1,\mathrm{y}}\sigma_{2,\mathrm{y}}) \tag{1}$$

where $\Delta_{0,j} \equiv \omega_{0,j} - \omega_a$, $\chi_j = g_j^2/\Delta_{0,j}$, $J = g_1 g_2 / 2\overline{\Delta}_0$, and the exchange coupling term has been written in terms of Bloch Pauli operators.

In this question, we look at how we can use this same cavity to carry out two-qubit dispersive measurements, and in particular *joint* (that is, entangled/entangling) two-qubit measurements.

To do this, we first add an external drive to the cavity, with coupling frequency (decay rate) κ. We assume that we are driving near the cavity resonance and hence also far detuned from the qubit frequencies, so we can ignore the effects of any cavity-mediated drives on the qubits themselves. Under these assumptions, the full driven system Hamiltonian becomes:

$$\frac{H}{\hbar} = -\sum_{j=1}^{2} \left[\frac{\Delta_{0,j} + 2\chi_j}{2} \right] \sigma_{\mathrm{j,z}} - \sum_{j=1}^{2} \chi_j a^\dagger a \sigma_{\mathrm{j,z}} + \frac{J}{2} (\sigma_{1,\mathrm{x}}\sigma_{2,\mathrm{x}} + \sigma_{1,\mathrm{y}}\sigma_{2,\mathrm{y}}) + i \left(a\Omega_{d_a}(t)^* - a^\dagger \Omega_{d_a}(t) \right) \tag{2}$$

where $\Omega_{d_a}(t) \equiv \sqrt{\kappa}\beta_{d_a} e^{-i\Delta_{d_a} t}$, β_{d_a} is the drive amplitude, and $\Delta_{d_a} \equiv \omega_{d_a} - \omega_a$.

As the system is driven, you will include the effects if dissipation through the following Heisenberg picture master equation:

$$\frac{d}{dt}O = -i[O, H] + \kappa \mathcal{D}^\dagger[a]O + \sum_{j=1}^{2} \gamma_{1,j} \mathcal{D}^\dagger[\sigma_{\mathrm{j,-}}]O + \sum_{j=1}^{2} \frac{\gamma_{\phi,j}}{2} \mathcal{D}^\dagger[\sigma_{\mathrm{j,z}}]O \tag{3}$$

For the remainder of the question, unless otherwise specified, you may assume that the following system parameters are fixed to $\omega_a/2\pi = 6.5$ GHz, $\Delta_{d_a}/2\pi = 0$ MHz, $g_1/2\pi = 60$ MHz, $g_2/2\pi = 80$ MHz, $\Delta_{0,1}/2\pi = -0.9$ GHz, $\Delta_{0,2}/2\pi = -1.6$ GHz, and $\kappa = 20$ MHz.

1. Under what conditions can you neglect the qubit-qubit exchange coupling term (i.e., set $J = 0$)? What is the relevant scaling ratio defining this scenario? **[1 mark]**

2. Assuming that you *can* neglect the exchange interaction ($J = 0$), derive the following dis-

persive equations of motion [**4 marks**]:

$$\frac{d}{dt}a = i\left(\sum_{j=1}^{2}\chi_j\sigma_{\mathrm{j,z}}\right)a - \frac{\kappa}{2}a - \Omega_{d_a}(t) \tag{4}$$

$$\frac{d}{dt}\sigma_{\mathrm{j,x}} = \left(\Delta_{0,j} + 2\chi_j(1+a^\dagger a)\right)\sigma_{\mathrm{j,y}} - \Gamma_j\sigma_{\mathrm{j,x}} \tag{5}$$

$$\frac{d}{dt}\sigma_{\mathrm{j,y}} = -\left(\Delta_{0,j} + 2\chi_j(1+a^\dagger a)\right)\sigma_{\mathrm{j,x}} - \Gamma_j\sigma_{\mathrm{j,y}} \tag{6}$$

$$\frac{d}{dt}\sigma_{\mathrm{j,z}} = \gamma_{1,j}\left(\sigma_{\mathrm{j,0}} - \sigma_{\mathrm{j,z}}\right) \tag{7}$$

where $\Gamma_j \equiv \frac{\gamma_{1,j}}{2} + \gamma_{\phi,j}$.

3. Consider now the case where the qubits are long-lived by comparison with any relevant cavity dynamics ($\gamma_{1,j} \ll \kappa$), so you may neglect the effects of qubit energy decay. Solve the cavity dynamics (find $a(t)$) for each two-qubit computational basis state ($|\psi\rangle = |j,k\rangle$ for $j,k \in \{g,e\}$). [**4 marks**]

4. Assuming a square drive pulse with a Rabi frequency $|\Omega_{d_a}|/2\pi = 5$ MHz and duration τ_m, starting at $t = 0$, create python code to plot the state-dependent cavity trajectories as a function of time. Attach your python code (jupyter notebook preferred, but not mandatory) and include your plots with your solutions. [**2 marks**]

5. Derive the cavity steady states for each qubit state combination, and calculate their overlap fidelities. [**2 marks**]

6. Show and discuss why the aim of this measurement design is that a cavity measurement will project the qubits onto a subspace of the form $|g,e\rangle\langle g,e| + |e,g\rangle\langle e,g|$ [**4 marks**].

   (a) How many different distinguishable measurement outcomes are there?

   (b) Describe the effect of the measurement on the qubit for each outcome.

   (c) What would the output state be for the measurement outcome corresponding to the subspace $|g,e\rangle\langle g,e| + |e,g\rangle\langle e,g|$, given an input qubit state of $|\psi\rangle = \frac{1}{2}(|g\rangle + |e\rangle) \otimes (|g\rangle + |e\rangle)$?

   (d) Outline what the full qubit measurement POVM set would look like. (You do not need to derive this rigorously. Please argue on physical grounds.)

   (e) What would be the effect of changing g_2 to $g_2/2\pi = 70$ MHz? Using only qubit frequency tuning, how could you re-calibrate the measurement setup to correctly measure the target subspace? Discuss what practical considerations you need to take into account when tuning up this system (e.g., to stay in the require operating regimes).

7. What drive Rabi frequency is required to reduce the overlap of the "non" overlapping outcomes to less than 1%? [**1 mark**]

8. Consider how this measurement might be affected if the Hamiltonian or dissipative terms we neglected are no longer negligible? [**2 marks**]